

Submit 4 problems total from the union of *LADW* Ch 6.3 & Ch 9.1-2 (copied below) and the supplementary problems. You are also welcome to revisit and submit problems from Ch 6.1-2, which were on HW6.

Please label all problems. Please put a “G” in front of the two problems you would like graded for correctness. Additionally, put an “F” in front of either of the problems graded for completion on which you would like written feedback from the grader. Do not submit any of the problems marked with an X. (None are!)

Suggested *LADW* problems: 6.3.1, 6.3.7, 6.3.12, 9.1.1, 9.2.1

Section 6.3

Exercises.

3.1. Show that the number of non-zero singular values of a matrix A coincides with its rank.

3.2. Find Schmidt decompositions $A = \sum_{k=1}^r s_k \mathbf{w}_k \mathbf{v}_k^*$ for the following matrices A :

$$\begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix}, \quad \begin{pmatrix} 7 & 1 \\ 0 & 0 \\ 5 & 5 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{pmatrix}.$$

3.3. Let A be an invertible matrix, and let $A = W\Sigma V^*$ be its singular value decomposition. Find a singular value decomposition for A^* and A^{-1} .

3.4. Find singular value decomposition $A = W\Sigma V^*$ where V and W are unitary matrices for the following matrices:

$$\begin{aligned} \text{a) } A &= \begin{pmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{pmatrix}; \\ \text{b) } A &= \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}. \end{aligned}$$

3.5. Find singular value decomposition of the matrix

$$A = \begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix}$$

Use it to find

- $\max_{\|\mathbf{x}\| \leq 1} \|\mathbf{A}\mathbf{x}\|$ and the vectors where the maximum is attained;
- $\min_{\|\mathbf{x}\| = 1} \|\mathbf{A}\mathbf{x}\|$ and the vectors where the minimum is attained;
- the image $A(B)$ of the closed unit ball in $\mathbb{R}^2$, $B = \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\| \leq 1\}$. Describe $A(B)$ geometrically.

3.6. Show that for a square matrix A , $|\det A| = \det |A|$.

3.7. True or false

- Singular values of a matrix are also eigenvalues of the matrix.
- Singular values of a matrix A are eigenvalues of A^*A .
- If s is a singular value of a matrix A and c is a scalar, then $|c|s$ is a singular value of cA .
- The singular values of any linear operator are non-negative.
- Singular values of a self-adjoint matrix coincide with its eigenvalues.

3.8. Let A be an $m \times n$ matrix. Prove that *non-zero* eigenvalues of the matrices A^*A and AA^* (counting multiplicities) coincide.

Can you say when zero eigenvalue of A^*A and zero eigenvalue of AA^* have the same multiplicity?

3.9. Let s be the largest singular value of an operator A , and let λ be the eigenvalue of A with largest absolute value. Show that $|\lambda| \leq s$.

3.10. Show that the rank of a matrix is the number of its non-zero singular values (counting multiplicities).

3.11. Show that the operator norm of a matrix A coincides with its Frobenius norm if and only if the matrix has rank one. **Hint:** The previous problem might help.

3.12. For the matrix A

$$A = \begin{pmatrix} 2 & -3 \\ 0 & 2 \end{pmatrix},$$

describe the inverse image of the unit ball, i.e. the set of all $\mathbf{x} \in \mathbb{R}^2$ such that $\|\mathbf{A}\mathbf{x}\| \leq 1$. Use singular value decomposition.

Section 9.1

Exercises.

1.1 (Cayley–Hamilton Theorem for diagonalizable matrices). As discussed in the above section, the Cayley–Hamilton theorem states that if A is a square matrix, and

$$p(\lambda) = \det(A - \lambda I) = \sum_{k=0}^n c_k \lambda^k$$

is its characteristic polynomial, then $p(A) := \sum_{k=0}^n c_k A^k = \mathbf{0}$ (we assuming, that by definition $A^0 = I$).

Prove this theorem for the special case when A is similar to a diagonal matrix, $A = SDS^{-1}$.

Hint: If $D = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\}$ and p is any polynomial, can you compute $p(D)$? What about $p(A)$?

Section 9.2

Exercises.

2.1. An operator A is called *nilpotent* if $A^k = \mathbf{0}$ for some k . Prove that if A is nilpotent, then $\sigma(A) = \{0\}$ (i.e. that 0 is the only eigenvalue of A).

Can you do it without using the spectral mapping theorem?

Supplementary problems for HW7

I. A property that a square matrix A can have is that there exists a power $k > 1$ such that $A^k = A$. Which matrices in $M_{2 \times 2}^{\mathbb{Z}/5\mathbb{Z}}$ have this property? Try to characterize them. How many are there? If there are matrices in $M_{2 \times 2}^{\mathbb{Z}/5\mathbb{Z}}$ that fail to have this property, explain how they behave differently.

II. Always, sometimes, or never? Prove if always or never, give an example each way if sometimes. For the following problems, the underlying vector spaces are over either $\mathbb{R}$ or $\mathbb{C}$.

- A matrix A has a square, A^2 .
- An operator A has a square, A^2 .
- An operator A has an inverse, A^{-1} .
- An operator A has a square root B , such that $B^2 = A$.
- If an operator A has a square root B , such that $B^2 = A$, then this square root is unique.

III. a) Show that in $\mathbb{R}^2$, $u \cdot v = \|u\| \|v\| \cos \theta$. Here by definition $u \cdot v = u_1 v_1 + u_2 v_2$.

b) Now carry out your proof so that it also works in $\mathbb{R}^n$.